

BREAK INTO AEROSPACE

Aircraft Performance Engineering

Subdomain Interview Preparation Guide

Targeted for the 2025–2026 Recruitment Cycle

Aircraft performance engineering is one of the largest and most consistently staffed disciplines in commercial aviation — and one of the least visible from outside the industry. Every airline with more than a handful of aircraft employs performance engineers. Every OEM employs them by the hundred. Every regulatory authority has them. And yet almost no university programme prepares students specifically for this work, which means that the field's technical vocabulary, its regulatory framework, and its practical methods are almost entirely learned on the job — or, for the candidate who prepares deliberately, before the interview.

The discipline sits at the intersection of aerodynamics, thermodynamics, atmospheric physics, regulatory compliance, and operational aviation. Its outputs are the numbers that determine whether an aircraft can legally depart from a given runway on a given day with a given load: the decision speed, the rotation speed, the takeoff field length, the obstacle clearance gradient, the climb ceiling, the en-route fuel burn, the landing distance with failed systems, the alternate fuel reserve. These are not academic quantities. They appear in the Approved Flight Manual, in the Flight Management System, in the Operational Flight Plan, and in the load sheet handed to the captain before every departure. When they are wrong, people die.

The employer landscape spans three distinct domains. Airlines — from regional carriers with 20 aircraft to the largest network carriers with 900 — employ performance engineers in fleet performance teams responsible for AFM data management, FMS database validation, performance software management, and operational efficiency. OEM performance groups — Airbus Performance Engineering and Flight Operations Support, Boeing Performance Engineering, Embraer Performance, Bombardier performance — develop the AFM data itself, conduct certification flight tests, and support airlines globally. A third domain, smaller but technically demanding, includes regulator-adjacent roles at EASA Flight Standards, FAA Aircraft Certification, and national airworthiness authorities, as well as specialist consultancies (AerData, Lido/Lufthansa Systems, Jeppesen) that build the performance calculation software that airlines and operators depend on.

The technical baseline is shared across all three domains. The regulatory framework — primarily CS-25 / FAR Part 25 for large aeroplanes — is the same. The physics of balanced field length, obstacle clearance, and climb gradient is the same. The FMS performance database format (the ARINC 702A standard) is the same. The cultural register, interview format, and career trajectory differ significantly between an airline performance team, an OEM certification group, and a performance software house.

PART I · WHAT AIRCRAFT PERFORMANCE ENGINEERING ACTUALLY LOOKS FOR

By Level

- **Intern / Graduate** — At entry level, performance engineering employers are not primarily screening for regulatory knowledge or FMS database expertise — both of which are learned in the first year on the job. They are screening for two things: quantitative fluency with the physics of flight, and the ability to translate a regulatory requirement into an engineering calculation.

The physics test at graduate level is applied, not theoretical. Can the candidate explain why a wet runway requires a longer accelerate-stop distance, and by how much? Can they explain why climb gradient is expressed as a percentage rather than an angle, and what that percentage must be for a

given obstacle? Can they estimate the fuel burn penalty of a 1,000 kg payload increase on a 2,000 nm stage? These questions do not require specialist knowledge — they require an understanding of drag, thrust, weight, and atmospheric conditions that any aerospace engineering graduate should have. The candidates who pass are those who have connected their aerodynamics and propulsion coursework to the operational reality of a commercial aircraft.

Graduate interviews at OEM groups (Airbus PEF in Toulouse, Boeing in Seattle or Everett, Embraer in São José dos Campos) typically involve a structured technical screen covering aerodynamics fundamentals, understanding of the certification framework, and at least one worked numerical problem — often involving climb gradient, takeoff field length estimation, or fuel planning. Airline performance team interviews vary more widely: a major network carrier may conduct a multi-stage process with technical testing and panel interviews; a regional carrier may have a more direct conversation with the chief performance engineer. In both cases, the candidate who has read and understood at least the performance sections of a public AFM — and can explain what a $V_1/V_R/V_2$ table is telling an operator — is immediately differentiated.

- **Mid-Level (Engineer, Senior Engineer, ~3–8 years)** — At mid-level, performance engineering interviews become operationally specific. The interviewer is assessing whether you can handle the actual work: managing an AFM revision, validating a performance software output against flight test data, investigating a performance discrepancy reported from line operations, or designing the performance methodology for a new aircraft type entering service at the airline.

Technical interviews at mid-level commonly involve regulatory discussion — which CS-25 amendment applies to which variant of which aircraft, what the ETOPS authorisation process requires in terms of performance demonstration, how the climb gradient requirements differ between a two-engine and four-engine aircraft, what a dry/wet/contaminated runway performance factoring methodology requires under EU-OPS / EASA Air Ops. OEM mid-level interviews go deeper into the certification methodology: how balanced field length is computed, what the net flight path requirements mean for obstacle clearance analysis, how performance deterioration with engine wear is modelled in the AFM.

The specific context driving mid-level hiring in 2025–2026: major airline fleet renewal programmes (Airbus A321XLR entry into service, Boeing 737 MAX continued expansion, Airbus A220 growth with new operators) are creating demand for performance engineers who can manage new type entry into service, including AFM data loading, FMS database validation, and SOP development for performance-limited operations. The retrofit of winglets and sharklets on existing fleets creates performance update work that requires re-validation of FMS databases. The expansion of ETOPS authorisation to new operators and longer diversion time authorities creates regulatory compliance work that is specifically performance-engineering driven.

- **Senior (Principal Engineer, Technical Lead, Chief Performance Engineer)** — At senior level, the interview is fundamentally about technical authority and organisational ownership. The senior performance engineer at an airline is the accountable person for the accuracy of the performance data that goes into the Flight Management System and the operational flight plan. The senior performance engineer at an OEM is the accountable person for the net flight path data in the AFM. Both roles require the ability to make technically defensible decisions under regulatory scrutiny, to manage the interface with the airworthiness authority, and to represent the organisation's technical position when an authority questions a methodology.

The senior interview typically involves a discussion of how you have managed a specific technical disagreement with a regulatory authority, how you have structured the evidence package for a new performance approval, or how you have diagnosed and resolved a systematic performance discrepancy that affected the safety case. The question “what is the most technically complex performance problem you have personally resolved, and how did you structure the evidence that your solution was correct?” is a live senior interview question in this field, and it has no short answer.

Hands-On Experience and Domain Knowledge Treated as Strong Signals

Domain	Why It Signals Depth
CS-25 / FAR Part 25 Requirements	The certification requirements for large aeroplane performance live in CS-25 Subpart B (CS-25.101 through CS-25.253) and the equivalent FAR Part 25. Engineers who have read these regulations and can explain what CS-25.121 requires for takeoff climb gradient, what CS-25.125 requires for landing distance, and what the 15% landing distance factor in EU Air Ops means are demonstrating baseline regulatory literacy.
Balanced Field Length	The fundamental performance calculation for commercial aircraft takeoff. Understanding how V_1 is optimised to minimise the balanced field length, what the obstacle-limited or tyre-speed limited cases are, and how runway slope and wind component affect the calculation is required mid-level domain knowledge.
Net Flight Path and Obstacle Clearance	Following an engine failure, a twin-engine aircraft must demonstrate a net flight path clearing all obstacles. Understanding the gross vs. net flight path distinction, the obstacle accountability area, and the methodology for area navigation (RNAV) departures are specifics OEM teams and airline engineers work with daily.
FMS Performance Database	The database contains the aerodynamic and propulsion model of the aircraft. Engineers who understand the ARINC 702A data format, validation against flight test/AFM data, failure mode management, and configuration management obligations demonstrate specifically valued operational performance depth.
ETOPS	ETOPS authorisation requires demonstration that the aircraft can reach an alternate aerodrome with required fuel reserves on one engine. Computing the critical fuel scenario and demonstrating that the aircraft's performance supports the diversion time authority demonstrates advanced operational engagement.
Contaminated Runway Performance	Runway contamination affects stopping distances and accelerate-go performance. Understanding why contaminated data is not fully certified, what braking action categories mean (friction coefficients), and how operational factoring methodologies apply distinguishes practical engineers from theoretical ones.
Performance Monitoring	Airlines track in-service performance to detect deterioration (increased fuel burn, aerodynamic degradation). Engineers who have built/maintained monitoring systems, extracting parameters from QAR data to separate deterioration from atmospheric uncertainty, demonstrate highly valued applied depth.
Flight Test Data Reduction	OEM groups develop AFM data from flight tests. Converting raw measurements into certified figures requires correction for non-standard atmospheres, weight, and configuration. Experience with standard correction methods (ESDU, AC 25-7D) demonstrates familiarity with the production process of performance data.
Atmospheric Modelling	Every performance parameter is a function of atmospheric conditions. Understanding how ISA deviations limit thrust/lift, how pressure altitude differs from indicated, and conservative wind assumptions are foundational literacy requirements for performance work.
Performance Software Tools	Commercial software (LESS, APM, Piano-X, Eido, Jeppesen EliteDeck Pro) is the daily tool. Familiarity with these tools, even academically (e.g., Piano-X), demonstrates practical engagement with the operational side of the discipline.

How Aircraft Performance Engineering Interviews Differ from Generic Aerospace Roles

- **The regulatory framework is load-bearing, not background.** In most aerospace engineering disciplines, regulatory awareness is helpful context. In aircraft performance engineering, it is the primary design constraint. The question “what field length does this aircraft need?” cannot be answered without specifying the regulatory standard, runway condition, and operational approval. Candidates who treat regulations as an afterthought demonstrate a misunderstanding of the engineering challenge.
- **Numerical estimation under uncertainty is a core competency.** Performance engineers are constantly called on to produce conservative estimates from imperfect inputs — estimating landing distance adjustments for braking action downgrades or available climb gradients given an engine failure. Interviewers probe this by asking candidates to structure a scenario and identify what data is required to declare a flight legally operable.
- **The interface between performance engineering and operations is real and tested.** Performance engineers work directly with flight crew, dispatchers, maintenance, and airworthiness authorities. Questions from crew often hide engineering issues. Interviewers probe whether candidates understand the operational chain — from OEM to AFM to FMS to load sheet — because an engineer who doesn’t understand it cannot identify where errors enter.
- **The FMS database is the performance engineer’s primary deliverable at airlines — and most graduates know nothing about it.** University programmes cover aerodynamics, not the ARINC 702A standard or airline configuration management obligations. A candidate who has read about FMS database structures demonstrates engagement that the majority of graduates lack.
- **Mistakes in this discipline have direct safety consequences — and interviewers know it.** An arithmetic error or misapplied regulatory requirement can contribute directly to a runway overrun or a terrain collision. Interviewers assess a candidate’s attitude toward rigour, verification, and uncertainty. A traceable, conservative verification process is more valuable than just getting the right answer.
- **Performance software literacy is assessed practically.** Interviews commonly include a practical element: working through a sample performance calculation using tools the organisation uses or validating a figure against AFM data. Candidates who have never used calculation tools will struggle with this practical component.

PART II · 20 SPECIFIC PREPARATION TASKS

Intern / Graduate

1. Work through a takeoff performance calculation from first principles — balanced field length, V_L , V_R , V_2 6–8 h

Skill: The foundational computation of aircraft performance engineering

Objective: Takeoff performance is the entry-level competency that every performance engineering interview probes. A candidate who can explain what balanced field length is — and work through a simplified calculation of accelerate-go distance and accelerate-stop distance to find the V_1 that equalises them — is demonstrating the physical understanding that underpins every subsequent performance topic. The calculation is not simple: it requires integrating thrust-minus-drag over the ground roll, accounting for the transition from ground to climb, computing obstacle clearance gradients, and understanding how temperature, pressure altitude, weight, and wind enter each term. The candidate who has done this — even with simplified assumptions, even in a spreadsheet — understands takeoff performance.

Done Signal: You have worked through a simplified balanced field length calculation — either from

a textbook (Torenbeek, Raymer, or equivalent) or from a derivation you have written yourself — for a twin-engine aircraft at a specified weight, temperature, and altitude. You can explain what happens to the balanced field length if the temperature increases by 15°C, if the runway elevation increases by 1,000 feet, or if the aircraft weight increases by 5,000 kg. You can define V_1 , V_R , and V_2 precisely and explain what constrains each in a limiting case.

6–8 h

2. Read and understand the performance sections of a publicly available Approved Flight Manual

4–5 h

Skill: Direct engagement with the primary regulatory document of commercial aviation performance

Objective: The AFM is the document that authorises the aircraft to operate — it is the OEM's certified statement of what the aircraft can do under specified conditions. Performance engineers at airlines manage the AFM; performance engineers at OEMs produce it. Yet most graduates have never seen one. AFMs for many aircraft types are publicly available (e.g., Boeing 737 NG AFM sections in accident reports, Airbus FCOMs). A candidate who has read the takeoff performance section of a real AFM — understood the table structure, the limiting cases, the footnotes, the applicability conditions — and can describe what they learned is demonstrating a specific, verifiable engagement with the discipline's primary output.

Done Signal: You have read the takeoff performance section of a real AFM or FCOM for a commercial transport aircraft. You can describe the table structure — what the rows and columns represent, what the reference condition is, how corrections are applied for actual conditions — and explain at least two notes or limiting cases that affected your understanding of what the table is and is not telling you. You can explain the difference between AFM data and Operations Manual performance data.

3. Understand the International Standard Atmosphere and its effect on every performance parameter

3–4 h

Skill: Atmospheric physics as the universal performance engineering context

Objective: Every aircraft performance parameter is a function of atmospheric conditions. Engine thrust is proportional to air density. Aerodynamic forces are proportional to dynamic pressure. Density altitude — the altitude in the ISA at which the local air density would be found — is the single number that integrates temperature and pressure into a performance-relevant quantity. Performance engineers who cannot reason fluently about ISA deviations — who cannot estimate the thrust penalty of a 20°C above-ISA day, or explain why a hot-and-high airport is doubly performance-limiting — are operating without their primary analytical tool.

Done Signal: You can derive the ISA lapse rate and explain why it is 6.5°C/km below the tropopause and 0°C/km above it. You can compute the pressure altitude at a given QNH and field elevation, the density altitude at a given pressure altitude and OAT, and the true airspeed at a given calibrated airspeed and pressure altitude. You can explain why an ISA+20°C day at 8,000 ft elevation can reduce takeoff thrust by more than 20% relative to ISA sea level, and why this has a compounding effect on climb gradient.

4. Prepare behavioural answers that connect technical rigour to operational consequence

2–3 h

Skill: STAR-format stories with the conservative-judgement emphasis performance engineering interviews demand

Objective: Performance engineering interviewers specifically probe how a candidate behaves when there is pressure to accept a performance result that does not feel right, when a captain wants an

answer the data does not clearly support, or when a database change has been made and there is uncertainty about whether the validation was complete. The stories that are strongest end with “the calculation said it was within limits but the margin was smaller than I expected... so I requested a hold while I checked the fuel quantity input.” Conservative judgement exercised specifically, with a traceable process, is what these stories must demonstrate.

2–3 h

Done Signal: You have four STAR stories covering: a situation where you caught a numerical error before it reached an operational consequence; a time you worked across organisational boundaries (engineering and operations); a situation where the answer was technically within limits but you judged further investigation was warranted; and a time you had to explain a complex technical constraint to a non-technical stakeholder under pressure.

5. Understand climb performance requirements — the gradient requirements of CS-25.121 and their operational implications

3–4 h

Skill: The regulatory requirements that govern aircraft operation with one engine inoperative

Objective: The four climb segments of CS-25.121 define the minimum climb capability required of a transport aircraft at various stages of the takeoff and initial climb with one engine inoperative. Understanding what the first segment (gear-down), second segment (gear up), final segment (cruise configuration) and en-route requirements are — and what happens if an aircraft cannot meet the second segment gradient requirement — explains why aircraft have climb-limited takeoff weights, why flex temperature de-rating is used, and why some runways are field-length-limited rather than climb-limited.

Done Signal: You can state the climb gradient requirements for each of the four takeoff flight path segments under CS-25.121 for a twin-engine aircraft, and explain what configuration, thrust rating, and airspeed applies to each. You can explain what “climb-limited takeoff weight” means, how it is determined, and why the second segment is most commonly the limiting segment for twin-engine jets at high temperatures and altitudes.

6. Get hands-on with a performance calculation tool — Piano-X, ACARS performance, or a real AFM dataset

5–7 h

Skill: Practical performance analysis capability before entering the industry

Objective: Piano-X is a publicly available aerodynamic performance analysis tool used widely to analyse aircraft performance. Using it to model a specific aircraft type demonstrates a practical engagement with performance analysis. The alternative is to work through a set of AFM tables manually: pick a specific runway at a specific airport (London City EGLC, Innsbruck LOWI), specify a realistic weight and OAT, and work through the published performance tables to determine the field length, gradient, and speed schedule.

Done Signal: You have used Piano-X or an equivalent tool to analyse the performance of a specific aircraft type across at least three scenarios (e.g. MTOW at ISA sea level, MLW at ISA+20°C 5,000 ft). Alternatively, you have manually worked through AFM performance tables for a specific airport and weather condition, arriving at a complete set of takeoff performance figures. You can explain every figure and identify which constraint is binding and why.

Mid-Level (~3–8 years)

All graduate tasks still apply. Behavioural stories must come from professional contexts involving real aircraft, real AFM data, or real operational decisions with traceable consequences.

7. Build a thorough working knowledge of ETOPS — the performance requirements, the critical fuel scenario, and the authorisation process 5–6 h

Skill: The performance methodology for extended-range operations — the most complex routine performance engineering task at airlines

Objective: ETOPS requires demonstration that the aircraft can reach the most critical diversion aerodrome with required fuel reserves while operating on one engine at a single-engine cruise altitude, having experienced a simultaneous pressurisation failure. The critical fuel scenario — the combination of diversion point, diversion weather, engine-out cruise altitude, pressurisation failure descent, and holding fuel — is computed for every ETOPS route. Engineers who have worked through this calculation understand what makes an ETOPS route performance-limiting and how the ETOPS fuel contingency is computed.

Done Signal: You can describe the critical fuel scenario for an ETOPS operation — the sequence of events modelled, the performance assumptions applied, and how the critical fuel reserve is derived. You can explain what makes a diversion aerodrome “adequate” and describe what would cause an ETOPS route to become non-compliant. You can name the relevant regulatory references (EASA Air Ops ORO.FC.145, FAA AC 120-42B).

8. Understand contaminated runway performance — methodology, regulatory basis, and how it differs from certified AFM data 4–5 h

Skill: Operational performance in degraded surface conditions — a daily reality in winter operations

Objective: Contaminated runway performance data is not certified performance in the CS-25 sense: it is performance data derived from a methodology that models the effect of contaminants on the ground roll and stopping distances. Understanding why a “demonstrated” stopping distance on a wet runway is not the same as a “factored” stopping distance under the AFM wet runway methodology, and what the operational procedures for different braking action conditions require, is vital domain knowledge.

Done Signal: You can explain the difference between the AFM wet runway landing distance and a “demonstrated” wet runway landing distance, and why operational factoring requirements apply additional factors. You can describe what the RCAM reports, how braking action categories relate to ICAO friction values, and what performance adjustment an operator must make. You can explain aquaplaning and its performance implication.

9. Understand FMS performance database structure — what it contains, how it is validated, and what goes wrong 4–5 h

Skill: The airline performance engineer’s primary operational deliverable

Objective: The FMS performance database is the airline performance engineer’s most important ongoing responsibility. Every AFM revision or aircraft modification may require a database update — and every update requires validation. Understanding what the ARINC 702A database contains, how it is produced, what the airline’s obligations are for validating it before use, and what the failure modes are when database errors are introduced (e.g., incorrect ETOPS fuel calculation) is critical.

Done Signal: You can describe what data is contained in a typical FMS performance database, how this data is derived from the AFM, and what a typical database validation process involves (ground testing, flight testing). You can describe at least one failure mode that has caused an FMS database error to affect operations and explain what the detection mechanism was.

10. Prepare a story about diagnosing a performance discrepancy in real operational data

2 h

Skill: Applied analytical judgement in a safety-critical operational environment

Objective: Performance discrepancies — an aircraft burning more fuel than predicted, a braking distance shorter than expected — are daily diagnostic challenges. They arise from data errors, measurement errors, modelling errors, or genuine deterioration. The engineer must determine the cause, distinguishing systematic issues from statistical noise. Engineers who have diagnosed a real discrepancy using operational flight data and AFM comparisons demonstrate applied analytical competence.

Done Signal: You have a story about a performance discrepancy you identified and investigated using real data. The story names the discrepancy, the data you examined, the hypotheses you formed, the analysis you conducted to distinguish between them, and the conclusion you reached. It includes at least one specific numerical comparison between predicted and observed performance.

11. Understand the net flight path and obstacle clearance methodology — including area navigation departures

4–5 h

Skill: The performance methodology that determines whether a takeoff is safe after engine failure

Objective: Following an engine failure at V_1 , the aircraft must follow a net flight path that clears all obstacles. The methodology — defining the obstacle accountability area, applying lateral clearance margins, computing the net flight path from the gross flight path by subtracting the regulatory gradient decrement — is technically demanding. The expansion of RNAV departures has complicated this, as the obstacle clearance area is narrower but navigation integrity must be demonstrated.

Done Signal: You can explain what the obstacle accountability area is, how its width is determined, and what the required clearance margin is. You can describe how the net flight path is derived from the gross flight path and what the regulatory gradient decrement is under CS-25. You can describe how an RNAV departure changes the obstacle accountability area.

12. Build working knowledge of aircraft performance monitoring — cruise performance analysis using flight data

4–5 h

Skill: The in-service tool for detecting performance deterioration and database errors

Objective: Airlines monitor cruise performance against certified AFM data to detect deterioration from engine wear, aerodynamic degradation, or systematic FMS database errors. The analysis typically extracts cruise segments from flight data, corrects for weight and atmospheric conditions, and compares the specific air range against the AFM prediction. Engineers must separate genuine deterioration from weight uncertainty and atmospheric variation.

Done Signal: You can describe the methodology for extracting and analysing cruise performance from operational flight data. You can explain what the expected uncertainty range is for a single cruise segment versus an average over 50 segments, and how you would distinguish genuine deterioration from a change in operational practices. You can explain what a 1% drag increase corresponds to in terms of annual fuel cost.

13. Prepare for a worked performance scenario under interview conditions — weight-limited takeoff from a constrained airport

5–7 h

Skill: Structured performance problem-solving under time pressure and regulatory constraint

Objective: Mid-level and senior interviews commonly include a practical scenario (e.g., Innsbruck LOWI Runway 26, OAT 25°C, 79,000 kg weight). This requires identifying the applicable perfor-

performance tables, applying correct corrections, checking multiple limiting constraints (field length, climb gradient, obstacle clearance), and arriving at a specific answer with a stated assumption set. 5–7 h

Done Signal: You have worked through at least two weight-limited takeoff performance calculations for performance-constrained airports using published AFM data or a performance tool. You know which constraint is binding, which limit is most sensitive to temperature, and what the regulatory basis is. You can set up and work through the calculation in under 20 minutes.

Senior Level (*Chief Performance Engineer, Principal Engineer, Technical Lead*)

All mid-level tasks still apply. Regulatory knowledge must extend to the AMC and GM level, not just the regulation text. ETOPS knowledge must include experience with a specific ETOPS authorisation process or route approval.

14. Prepare to walk through a complete AFM revision cycle — from OEM amendment through airline implementation to validation 4–6 h

Skill: End-to-end AFM configuration management at the level a senior performance engineer owns

Objective: The AFM revision cycle is the core operational process senior engineers own. The cycle has safety-critical quality gates at every step, from the OEM issuing an amendment through to the flight operations acceptance. The senior interview probes whether you have personally managed this process: assessing operational impact, driving the database update, validating against the amendment, and obtaining airworthiness authority approval.

Done Signal: You can walk through the complete AFM revision cycle for one specific amendment from your career — naming its content, the assessment process, implementation steps, validation methodology, and the approval obtained from your competent authority. You can identify the most significant risk in the cycle and describe the control you had in place for it.

15. Prepare a technically grounded position on one current development in performance engineering — electric propulsion integration, AAM performance certification, or GNSS-based performance monitoring 3–4 h

Skill: Forward-looking technical awareness at senior engineering depth

Objective: Performance engineering in 2025–2026 is encountering new aircraft types that don't fit traditional assumptions. Electric aircraft have throttle response and battery state-of-health dependencies. eVTOL vehicles are certified with fundamentally different performance requirements due to distributed electric propulsion. Senior engineers must engage deeply with what these transitions require, adapting performance methodologies to new failure modes and realities.

Done Signal: You have a prepared 3-minute technical discussion covering one specific engineering challenge: how battery state-of-health affects performance guarantees, what “balanced field length” means for a distributed electric propulsion eVTOL, or how GNSS-based monitoring could complement conventional methods. Your position engages with the specific engineering difficulty, not just high-level regulations.

16. Prepare to lead a performance engineering trade-off discussion — route planning, fleet selection, or ETOPS authorisation 4–5 h

Skill: Technical authority at the level of organisational decision-making

4–5 h

Objective: Senior engineers make or advise on decisions with direct commercial and safety consequences: whether a proposed route is performable, whether a new airport requires new performance approvals, or whether an ETOPS extension is achievable. These require integrating regulatory requirements, aircraft capability, constraints, and commercial objectives — and defending the conclusion.

Done Signal: You have worked through at least two senior performance trade-off problems — a proposed new route assessment and a fleet performance limitation scenario. You can state the performance constraints, data required, regulatory approvals needed, and timeline. You can do this conversationally, in under 8 minutes per scenario, without reference to notes.

3–4 h

17. Understand the airworthiness authority interface — how performance approvals are obtained, maintained, and audited

Skill: Regulatory authority management at the level of a senior performance engineer

Objective: Senior engineers are the primary technical interface with the competent authority. This means understanding what EASA or the FAA requires as evidence for a performance approval (ETOPS, special airport, contaminated runway). It means understanding why an analytical demonstration is not always sufficient, when a flight demonstration is required, and what the standard of evidence is for novel methodologies.

Done Signal: You can describe the content and structure of a performance approval package for one specific approval type, naming the regulatory basis, evidence required, and typical timeline. You can describe one instance where you managed such a package, what the authority's principal concerns were, and how those concerns were addressed.

4–5 h

18. Build quantitative estimation fluency for performance — takeoff weight limits, fuel burns, climb gradients — without looking up tables

Skill: Senior-speed first-principles reasoning about performance

Objective: Senior engineers are expected to sanity-check operational requests and tool outputs in real time. Estimating if a flight can take off from a specific runway at MTOW requires estimating field length at ISA sea level, penalties for temperature/altitude, and comparing against available runway. These estimates must be done with stated assumptions, quickly enough to be useful operationally.

Done Signal: You have worked through five first-principles performance estimation problems (e.g., field length estimation, second segment gradient estimation, specific air range and fuel burn, ETOPS critical fuel scenario). You state assumptions before numbers, arrive at order-of-magnitude answers with stated uncertainty, and identify the most load-bearing assumption in each.

All Levels

1–2 h

19. Prepare a specific answer to “Why performance engineering — and why this specific domain?” that reflects genuine technical engagement

Skill: Technically specific motivation that distinguishes candidates committed to the discipline

Objective: Performance engineering requires simultaneous fluency in aerodynamics, propulsion, atmospheric physics, and regulatory compliance. Candidates who explain their interest in generic terms are not demonstrating they understand what makes the discipline technically demanding. A strong answer engages with a specific technical challenge (e.g., ETOPS 370-minute extensions, hybrid-electric performance methodologies, RNAV obstacle clearance from high terrain).

Done Signal: You have a 90-second answer that names your target performance engineering do-

main, identifies one specific current engineering challenge in it, explains what makes it technically difficult, and connects your background to contributing to its solution. The answer cannot be generic. 1–2 h

20. Map your technical background to a specific employer type, aircraft type, and performance domain — and verify regulatory familiarity 1–2 h

Skill: Precise self-positioning in a field where employer type determines daily work more than in most engineering disciplines

Objective: The airline engineer and the OEM engineer share vocabulary but do different work. Airlines manage existing AFM data and FMS databases. OEMs develop new AFM data from flight tests. FMS database developers build translation software. Regulatory specialists audit methodologies. A candidate who cannot explain why their background positions them for a specific domain is asking the recruiter to do their work.

Done Signal: You can state in under 60 seconds: the employer type you are targeting, the aircraft type family most relevant to your experience, the specific performance domain you are most prepared for, your regulatory familiarity (CS-25 vs. FAR 25), and two specific technical contributions your background makes to the target role.

SUMMARY REFERENCE TABLE

#	Task Description	Target Level	Effort
1	Takeoff performance from first principles — balanced field length, $V_1/VR/V_2$	All	6–8 h
2	Read a real AFM performance section — table structure, limiting cases, footnotes	All	4–5 h
3	ISA and its effect on every performance parameter — quantitative fluency	All	3–4 h
4	STAR stories — error detection, conservative judgement, operational interface	All	2–3 h
5	CS-25.121 climb requirements — four segments, gradient values, limiting cases	All	3–4 h
6	Hands-on performance tool or AFM table calculation — constrained real scenario	All	5–7 h
7	ETOPS — critical fuel scenario, performance requirements, authorisation process	Mid+	5–6 h
8	Contaminated runway — methodology, braking action categories, regulatory basis	Mid+	4–5 h
9	FMS performance database — ARINC 702A structure, validation, failure modes	Mid+	4–5 h
10	Performance discrepancy diagnosis story — data-driven root cause	Mid+	2 h
11	Net flight path and obstacle clearance — gross/net distinction, RNAV departures	Mid+	4–5 h
12	Aircraft performance monitoring — cruise analysis from flight data	Mid+	4–5 h
13	Worked constrained-airport scenario under time pressure	Mid+	5–7 h
14	Complete AFM revision cycle story — amendment through validated implementation	Senior	4–6 h
15	One current development — electric aircraft, AAM, or GNSS-based monitoring	Senior	3–4 h
16	Performance trade-off discussion — route assessment or ETOPS extension	Senior	4–5 h
17	Airworthiness authority interface — approval package, audit, evidence standard	Senior	3–4 h
18	First-principles performance estimation — five problems, stated assumptions	Senior	4–5 h
19	“Why performance engineering — why this domain?” — technically specific	All	1–2 h
20	Employer type / aircraft type / performance domain self-positioning	All	1–2 h

THE STRUCTURAL REALITY

Aircraft performance engineering is one of the few disciplines where the engineer's output is used directly, every day, by people whose lives depend on it being correct — not through multiple layers of system integration and operational procedure, but directly. The number in the box on the load sheet is the performance engineer's number. The maximum takeoff weight limitation on the departure clearance is derived from the performance engineer's analysis. The fuel reserve the captain carries to the ETOPS alternate aerodrome was computed using a methodology the performance engineer validated. There is almost no other engineering discipline where the chain from the engineer's calculation to the operational consequence is this short.

This directness creates a culture that is genuinely different from other aviation engineering disciplines. Performance engineers are conservative by professional formation — not because they are risk-averse personally, but because the regulatory methodology they apply is explicitly designed to be conservative, and departing from that conservatism requires documented justification that will be reviewed by a regulator. The candidate who says "I'd look at that more carefully before approving it" is not being overly cautious — they are demonstrating an understanding of what the job actually requires.

The field also sits at an unusual intersection of the scientific and the operational. The physics of balanced field length is aerodynamics and thermodynamics. The regulatory requirements of CS-25.121 are administrative and legal. The operational implications of a climb-limited departure are flight operations and dispatch. The FMS database that implements all of the above is software engineering. The performance engineer who is comfortable in all four of these modes — who can move from a discussion of turbofan thrust modelling to a discussion of EU Air Ops regulatory compliance to a discussion with a dispatcher about why today's fuel load is higher than the flight plan — is the one who genuinely thrives in the role.

The engineers who struggle are those who treat performance engineering as a lookup discipline — who think the job is to find the right table in the AFM and read off the answer. Performance engineering at any serious level of practice is an analytical discipline. The tables are the output of an analytical process, and understanding that process — the physics, the regulation, the methodology — is the only way to know when a table applies, when it does not, and what to do when the situation falls outside the cases the table covers. Those cases arise in operations every day. They are where performance engineering earns its keep.